

Intrusion Methodology Analysis

Title: NET-2650 – Intrusion Methodology Analysis

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1. Intrusion Methodologies

One common intrusion method is credential abuse. This occurs when an attacker gains access to valid usernames and passwords, often through phishing, data breaches, or password reuse. Instead of hacking directly into systems, the attacker logs in as a legitimate user, making the activity harder to detect.

Another common method is brute force attacks. In this technique, attackers attempt many password combinations until they successfully guess the correct one. This can be done manually or through automated tools that rapidly try different credentials.

A third method is service misuse. This happens when attackers take advantage of legitimate services such as RDP, SSH, or file sharing systems. If these services are exposed or poorly secured, attackers can use them to gain access without exploiting a vulnerability.

2. Indicators of Compromise

For credential abuse, logs may show successful logins from unusual locations, times, or devices. For example, a user logging in from a different country or at an unusual hour could indicate compromised credentials. There may also be multiple login sessions occurring at the same time.

For brute force attacks, logs often show repeated failed login attempts followed by a successful login. These attempts may come from the same IP address or multiple IPs in a short period of time. Account lockouts may also occur if security controls are in place.

For service misuse, logs may show connections to services like RDP or SSH from unfamiliar IP addresses. There may also be unusual command activity, unexpected file access, or changes made to the system after login.

3. Analyst Reasoning

A security analyst must carefully distinguish between normal and suspicious activity. Not all unusual behavior is malicious. For example, a user logging in at a different time may simply be working late or traveling.

To avoid false positives, analysts compare current activity with known patterns, such as typical login times, locations, and user behavior. They may also correlate multiple logs, such as authentication logs and network logs, to confirm whether an event is truly suspicious.

Analysts also look for patterns instead of isolated events. A single failed login is normal, but repeated failures followed by a success are more concerning. By using context and multiple data sources, analysts can make more accurate decisions and reduce false alarms.

AI-Usage Log:

Tool | Prompt Summary | Output Used | Validation | Limitations ChatGPT | Help writing intrusion analysis | Draft structure and content | Reviewed and aligned with course material | May simplify complex concepts

References:

MITRE ATT&CK Framework Microsoft Learn NIST SP 800-92